

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
8	Indicative content Section A: From epidermis and across cortex: • Apoplast + symplast + vacuolar pathway • (Apoplast) Along cell walls • (Symplast) through cytoplasm/ plasmodesmata • (Vacuolar pathway) passes through tonoplast/ cytoplasm • Down water potential gradient / by osmosis Section B: Into vascular tissue: • Endodermis has Casparian strip • Waterproof / composed of suberin • Blocks apoplast pathway • Water {forced / passes} into symplast • Active transport of {salts / minerals} <u>into xylem</u> • Lowers water potential in <u>xylem</u> • Water enters <u>xylem</u> by osmosis / down water potential gradient Waterlogged soil: • {Less / no} oxygen (for respiration) • {Less / no} respiration so {less / no} {energy / ATP} • for active transport of minerals in the {endodermis/ root hairs} Must be in context of no oxygen • Lower water potential gradient as a result of less active transport of salts / minerals into xylem • Less water enters xylem which lowers root pressure	5	4				